SCI-BEAM
Engineering Conformance Specification

Generated engineering view of the shared normative core
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Draft status. This document contains no independent science. Normative equations, definitions, assumptions, requirements, and predictions are imported from the same shared v0.1 core as the Scientific Rationale. Conformance is not claimed here.

1 Conformance target

An implementation conforms only if it realizes the shared authority below and produces evidence for every applicable requirement and prediction without weakening upstream validity or upgrading scientific authority. Field names, schemas, algorithms, storage formats, and current behavior are intentionally unspecified.

Area	Required evidence class
Identity and state	Demonstrate immutable parent/source/observation identity, explicit occurrence-scoped detector binding, and one-way requested/effective/realized state.
Numerical operator	Demonstrate the declared source-convolved ellipse, amplitude normalization, background basis, admitted support, and objective, including edge and degeneracy behavior.
Validity and uncertainty	Demonstrate causal exclusions, typed unavailable states, covariance method/assumptions, nuisance treatment, and cause-preserving QC.
Prior and convergence	Demonstrate compatibility, blind fallback, common-objective candidate selection, influence record, and conjunctive termination state.
Publication	Demonstrate an atomic required bundle, explicit optional-parent relations, candidate non-promotion, and compatibility checks.
Authority gates	Label algebraic, implementation, response/representation, observational-performance, and production-readiness evidence separately.

2 Conformance procedure

For each requirement, evidence shall identify the realized policy and scientific identity, exercise applicable positive and negative cases, and compare the produced state and payload to the shared equation or invariant. A finite match without matching identity and validity is a failure. Skipped required inputs are not a pass. Threshold-dependent evidence shall report the supplied policy rather than imply a v0.1 default.

Prediction cases are falsifiable expectations of the shared operator. Synthetic cases can test algebra and state transitions; they cannot establish observational performance or production readiness. Correlated-noise calibration, response fidelity, and calibration promotion require evidence from their owning layers.

3 Normative notation

Symbol	Meaning
o, d, i	Observation, stable detector identity, and admitted sample or map-pixel index. Indices in memory are zero-based; persisted FITS/WCS pixels are one-based.
$\mathbf{x}_i = (x_i, y_i)^\mathsf{T}$	Externally supplied AltAz tangent-plane azimuth/elevation offset, in arcseconds, about the declared source. WCS supplies signs and handedness.
y_i	Admitted detector datum in declared signal unit U_y; the active map boundary is mJy/beam but that label alone proves neither absolute calibration nor complete response.
v_i	Upstream eligibility and validity state, evaluated before the numerical payload.
$\mathcal{I}_d$	Admitted support for detector d: indices satisfying upstream eligibility, finite-data policy, coordinate validity, and the effective support rule.
$\mathbf{C}_d$	Declared covariance of $\mathbf{y}_d$ on $\mathcal{I}_d$, or a typed approximation $\widetilde{\mathbf{C}}_d$.
$S(\mathbf{u}; \boldsymbol{\phi})$	Declared, non-negative calibrator surface-brightness shape with finite first moment, unit integral, and declared reference origin fixed by $\int \mathbf{u} S(\mathbf{u})\, d^2\mathbf{u} = \mathbf{0}$, with nuisance parameters $\boldsymbol{\phi}$.
$G(\mathbf{u}; \boldsymbol{\Sigma})$	Peak-normalized elliptical Gaussian beam.
$T(\mathbf{u}; \boldsymbol{\Sigma}, \boldsymbol{\phi})$	Source-convolved template normalized to unity at the declared source-reference origin, $T(\mathbf{0}) = 1$, as defined in Section 5; that origin need not be the profile maximum.
A_d	Fitted source excess at the translated declared reference origin, in U_y; it is neither necessarily the profile peak nor source flux, calibration, or sensitivity.
$\boldsymbol{\mu}_d = (\mu_{x,d}, \mu_{y,d})$	Fitted translation of the declared zero-first-moment source-reference origin in the admitted tangent plane, in arcseconds; equivalently the modeled convolved-profile centroid when the admitted moments exist.
$\sigma_{\text{maj}}, \sigma_{\text{min}}, \theta$	Gaussian major/minor standard deviations and orientation, with $\sigma_{\text{maj}} \geq \sigma_{\text{min}} > 0$ and $\theta \in [0, \pi)$.
$\mathbf{q}_d$	Model parameter vector $(A_d, \boldsymbol{\mu}_d, \sigma_{\text{maj}}, \sigma_{\text{min}}, \theta, \boldsymbol{\beta}_d)$.
$\boldsymbol{\beta}_d$	Coefficients of the declared bounded background basis.
Q_d	Declared generalized least-squares or likelihood objective.
$d_j(a, b)$	Declared stability metric for available parameter j; for orientation, $d_\pi(a, b) = \min_{n\in\mathbb{Z}} \lvert a - b + n\pi \rvert$.
$\mathcal{A}_d^{(k)}, c_d^{(k)}, \mathcal{V}^{(k)}$	Parameter-availability set, selected locator/measurement-candidate identity (including a typed none state), and valid-detector set at iteration k.
$F_{s,a}$	Immutable source-model flux quantity supplied for array a, with its epoch, passband, convention, and uncertainty.
K_d	Typed candidate conversion factor $F_{s,a}/A_d$ when dimensional and authority preconditions hold.

The words *shall* and *shall not* are normative. *May* marks an allowed option whose realized state must be recorded. “Unavailable” is a typed scientific state, never an undocumented numeric sentinel.

4 Definitions and state model

Estimand. For each attempted detector, SCI-BEAM estimates an observation-local, source-convolved response profile over admitted support. The base family is one elliptical Gaussian beam convolved with the separately declared calibrator brightness model, plus a declared constant or first-order planar background. The point-source case is the limiting distribution $S \to \delta$. More complex beam components are outside v0.1 unless a successor contract names their identifiability and normalization.

Ellipse convention. $+x$ is the WCS-declared azimuth-offset direction and $+y$ the elevation-offset direction. The orientation θ is measured counterclockwise in that coordinate plane from $+x$ to the major-axis eigenvector, modulo π.

Major/minor ordering removes axis-label degeneracy. At exact circularity, width remains defined and θ is unavailable, not zero.

Reference origin, centroid, and amplitude convention. The admitted source shape has unit integral and finite first moment, and its declared reference origin is fixed by zero first moment. Because the Gaussian is also centered, convolution preserves that reference centroid. The fitted $\boldsymbol{\mu}_d$ translates this declared origin into the admitted tangent plane; it is a relative modeled centroid, not absolute astrometry. The template is normalized by its positive value at that origin, so $T(\mathbf{0}) = 1$ exactly. Consequently A_d is the modeled source excess above background at $\mathbf{x} = \boldsymbol{\mu}_d$. Neither H nor T is required to attain its global maximum there for an asymmetric admitted source. A_d is not the intrinsic source flux, unconvolved beam peak, integrated flux, response coefficient, promoted calibration, or sensitivity.

Background. The effective policy chooses either a constant basis [1] or a plane $[1, x - x_\star, y - y_\star]$ about a recorded origin $\mathbf{x}_\star$. No higher-order or adaptive term is part of v0.1. A background coefficient is reportable only when the admitted support makes the joint source/background design identifiable.

Attempt and terminal states. Every detector bound to the observation has an attempt record. The ordered processing states are `not-attempted`, `candidate-located`, `fitted`, and `optimizer-complete`; these do not imply one another when a prior stage fails. Exactly one terminal disposition is recorded: `valid`, `rejected`, `invalid-input`, `failed`, `non-converged`, or `unavailable`, with one or more cause codes. Missing, disabled, unsupported, and unavailable inputs remain distinguishable causes.

Authority layers. Algebraic contract correctness means the equations and identities are internally satisfied. Implementation conformance means an implementation realizes those equations and states. Representation/response fidelity means the admitted coordinates, support, transfer response, source model, and covariance represent the intended observation. Observational performance means bias and uncertainty have been measured on appropriate data. Production readiness additionally requires owner-approved thresholds, operations, and evidence. No layer implies a later layer.

Atomic result. One observation-local publication transaction binds the immutable observation/source identity, requested/effective/realized policy, detector identity relation, fit state and parameters, covariance statement, prior route and influence, convergence trace summary, QC causes, calibration-candidate state, and provenance. Optional map or TOD companions are views with explicit parent identity; they cannot replace the required result/QC/provenance records.

Compatibility. Two artifacts are compatible only after explicit comparison of contract version, model/normalization identity, observation and source identity, detector binding scope, frame/WCS identity, units, response status, source-model convention, covariance status, and validity. Equal filenames, row counts, row order, local UID text, or finite payloads are insufficient.

5 Forward model and objective

Let

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \tag{1}$$

$$\boldsymbol{\Sigma} = \mathbf{R}(\theta)\,\mathrm{diag}(\sigma_{\mathrm{maj}}^2, \sigma_{\mathrm{min}}^2)\,\mathbf{R}(\theta)^\mathsf{T}, \tag{2}$$

$$G(\mathbf{u}; \boldsymbol{\Sigma}) = \exp\!\left(-\tfrac{1}{2}\mathbf{u}^\mathsf{T}\boldsymbol{\Sigma}^{-1}\mathbf{u}\right), \tag{3}$$

$$H(\mathbf{u}; \boldsymbol{\Sigma}, \boldsymbol{\phi}) = \int S(\mathbf{t}; \boldsymbol{\phi}) G(\mathbf{u} - \mathbf{t}; \boldsymbol{\Sigma})\, d^2\mathbf{t}, \tag{4}$$

$$T(\mathbf{u}; \boldsymbol{\Sigma}, \boldsymbol{\phi}) = \frac{H(\mathbf{u}; \boldsymbol{\Sigma}, \boldsymbol{\phi})}{H(\mathbf{0}; \boldsymbol{\Sigma}, \boldsymbol{\phi})}. \tag{5}$$

The admitted source model shall have finite first moment and satisfy

$$\int S(\mathbf{u}; \boldsymbol{\phi})\, d^2\mathbf{u} = 1, \qquad \int \mathbf{u} S(\mathbf{u}; \boldsymbol{\phi})\, d^2\mathbf{u} = \mathbf{0}, \qquad H(\mathbf{0}) > 0.$$

Thus the declared reference origin is the source centroid, convolution with the centered Gaussian preserves that centroid, and Equation 5 guarantees $T(\mathbf{0}) = 1$. It does not assert that $\mathbf{0}$ is the global maximum for an asymmetric source. The detector model is

$$m_i(\mathbf{q}_d, \boldsymbol{\phi}) = A_d T(\mathbf{x}_i - \boldsymbol{\mu}_d; \boldsymbol{\Sigma}_d, \boldsymbol{\phi}) + \mathbf{b}_i^\mathsf{T} \boldsymbol{\beta}_d, \tag{6}$$

where $\mathbf{b}_i$ is the effective constant or planar background basis.

For an admitted covariance, the estimator minimizes

$$Q_d = (\mathbf{y}_d - \mathbf{m}_d)^\mathsf{T} \mathbf{C}_d^{-1} (\mathbf{y}_d - \mathbf{m}_d) + \log |\mathbf{C}_d|, \tag{7}$$

up to additive constants. If only a declared diagonal or other approximation is supplied, the same expression uses $\widetilde{\mathbf{C}}_d$ and every covariance, standardized-residual, and significance claim is labeled conditional on that approximation. Singular covariance is handled by a declared, provenance-bearing factorization on an explicitly retained subspace; silent regularization is forbidden.

The fit covariance may be reported as a local likelihood covariance

$$\mathbf{V}_{q,d} = \left[\tfrac{1}{2} \nabla \nabla^\mathsf{T} Q_d(\widehat{\mathbf{q}}_d)\right]^{-1} \tag{8}$$

only when the retained parameterization is locally identifiable, the solution is interior for the reported parameters, and the likelihood/covariance assumptions used by the formula are declared. Otherwise it is unavailable or supplied by a separately identified valid procedure. With nuisance vector $\boldsymbol{\eta}$ and admitted covariance $\mathbf{V}_\eta$, first-order propagation is

$$\mathbf{V}_{\text{total}} = \mathbf{V}_q + \mathbf{J}_\eta \mathbf{V}_\eta \mathbf{J}_\eta^\mathsf{T} \tag{9}$$

with cross-terms included when dependence exists. Omitted calibrator, coordinate, response, atmosphere, bandpass, or calibration uncertainty is named; "total" is prohibited when a material term is unavailable.

When the admitted source flux $F_{s,a}$ and fitted coefficient A_d have compatible conventions and $A_d \neq 0$, BEAM may form

$$K_d = \frac{F_{s,a}}{A_d}, \qquad \mathrm{Var}(K_d) \simeq \frac{\mathrm{Var}(F_{s,a})}{A_d^2} + \frac{F_{s,a}^2 \mathrm{Var}(A_d)}{A_d^4} - 2\frac{F_{s,a} \mathrm{Cov}(F_{s,a}, A_d)}{A_d^3}. \tag{10}$$

If any required variance or dependence term is unknown, candidate uncertainty is unavailable rather than silently independent. K_d remains a typed candidate for SCI-CAL review.

An observation-local iteration may alternate locator and measurement phases. The effective policy declares a metric d_j and scale s_j for each parameter, objective scale s_Q, tolerances ϵ_j, ϵ_Q, support and valid-detector stability rules, and maximum iteration $N_{\max}$. Linear scalar parameters may use absolute difference only when the policy says so; orientation shall use the modulo-π distance

$$d_\pi(a, b) = \min_{n \in \mathbb{Z}} |a - b + n\pi|.$$

Let $\mathcal{A}_d^{(k)}$ be the parameter-availability set, $c_d^{(k)}$ the selected locator/measurement-candidate identity including a typed none state, and $\mathcal{V}^{(k)}$ the valid-detector set. Convergence occurs only when

$$\begin{aligned} \max_{j \in \mathcal{A}_d^{(k)} \cap \mathcal{A}_d^{(k-1)}} \frac{d_j(q_j^{(k)}, q_j^{(k-1)})}{s_j} &\leq \epsilon_j, \qquad & \mathcal{A}_d^{(k)} &= \mathcal{A}_d^{(k-1)}, \\ \frac{|Q^{(k)} - Q^{(k-1)}|}{s_Q} &\leq \epsilon_Q, \qquad & c_d^{(k)} &= c_d^{(k-1)}, \\ \mathcal{I}_d^{(k)} = \mathcal{I}_d^{(k-1)}, & \quad \mathcal{V}^{(k)} = \mathcal{V}^{(k-1)}. \end{aligned} \tag{11}$$

for the declared number of consecutive transitions and only where the objective is mutually defined. Availability equality is tested separately: a parameter unavailable at both transitions, including θ at circularity, contributes no numerical metric; a change into or out of availability fails stability. Candidate, support, and valid-detector stability are separately recorded components, not entries hidden in a raw parameter vector. No numerical tolerance or count is a universal scientific constant in v0.1.

6 Assumptions, dependencies, and unavailable claims

A1. The immutable source identity and per-array brightness/flux model are upstream inputs. Source selection and catalog estimation are not BEAM results.

A2. Coordinates and WCS are externally supplied in the declared AltAz tangent plane. Relative centroid inference does not establish timing, detector-coordinate truth, pointing correction, or absolute astrometry.

A3. Upstream validity is causal and precedes payload evaluation. BEAM adds fit-specific admission and validity without weakening upstream exclusions.

A4. The admitted map/signal unit and response status are preserved. Unless the complete transfer response is established upstream, fitted shape is a conditioned effective profile and an intrinsic detector-plus-telescope beam claim is unavailable.

A5. The calibrator model is non-negative, unit-integral, has a finite first moment with declared reference origin satisfying $\int \mathbf{u}S(\mathbf{u})\,d^2\mathbf{u} = \mathbf{0}$, is evaluated for the declared epoch/passband convention, and is accompanied by uncertainty or an explicit unavailable state.

A6. The base beam family is adequate only as a declared model. Good residuals or convergence do not prove that adequacy; model-inadequacy diagnostics remain distinct from optimizer failure.

A7. The covariance or approximation describes the retained support sufficiently for the statistical claims made. A diagonal approximation does not authorize calibrated uncertainty under correlated residuals.

A8. The support and background basis jointly identify source, centroid, widths, angle where non-circular, amplitude, and background. Parameters that are not identified are unavailable.

A9. A compatible TolAPT prior is soft locator initialization and bounded gating information only. It contributes no penalty to Equation 7, is not exact UID truth, and cannot veto the declared blind route.

A10. Internal iteration is confined to one immutable observation/result identity. Cross-observation learning, feedback, recurrence, and restart are FRUIT responsibilities.

A11. SCI-CAL alone may promote a calibration candidate. Sensitivity is unavailable as a BEAM-owned result in v0.1 because no exact joint convention for noise, time, atmosphere, calibration, and bandwidth is authorized.

A12. Production thresholds are policy inputs, not inferred physical constants. Their provenance, requested value, effective value, realized value, owner, and change identity are recorded.

Dependencies on TolProj/TolTECA photometry, SCI-CAL, ALIGN/AST, RTC/PTC, VAL, SCI-MAP, TolAPT, `toltec_beammap`, and FRUIT remain open at their stated interfaces. This contract does not certify those inputs or close their authority decisions.

7 Normative requirements

ID	Requirement
SCI-BEAM-REQ-001	The result shall bind immutable observation, source, array, detector, parent-input, contract-version, and realized-policy identities.
SCI-BEAM-REQ-002	Stable detector binding shall be explicit and occurrence-scoped; row order, slot, array, network, or local key text shall not substitute for it.
SCI-BEAM-REQ-003	Requested, effective, observation-resolved, and realized model/support/prior/iteration/QC state shall be distinct, immutable, and one-way.
SCI-BEAM-REQ-004	Coordinates shall be the supplied AltAz tangent-plane azimuth/elevation offsets in arcseconds, with WCS identity, signs, handedness, validity, and uncertainty state recorded.
SCI-BEAM-REQ-005	The source input shall declare immutable identity, epoch, passband/spectral convention, extent, flux/brightness normalization, uncertainty, and upstream authority.
SCI-BEAM-REQ-006	The signal/map input shall declare unit, response/kernel status, calibration lineage, support, validity, covariance status, provenance, and stable detector binding.
SCI-BEAM-REQ-007	Upstream-invalid samples or detectors shall be excluded before payload evaluation; missing, disabled, rejected, invalid, failed, and unavailable shall remain distinct.
SCI-BEAM-REQ-008	The v0.1 forward model shall be Equation 6: a declared finite-source or point-limit template convolved with one elliptical Gaussian plus a constant or planar background.
SCI-BEAM-REQ-009	The ellipse shall use ordered positive axes, angle measured counterclockwise from WCS $+x$, period π, and unavailable angle at exact circularity.
SCI-BEAM-REQ-010	The template shall use Equation 5; the source shall satisfy the declared zero-first-moment reference convention, and the $T(\mathbf{0}) = 1$ reference-origin normalization and source-model identity shall accompany every amplitude.
SCI-BEAM-REQ-011	A_d shall mean modeled source excess at the translated declared reference origin in the admitted signal unit; it shall not be labeled a profile peak unless separately established, nor relabeled as intrinsic flux, response, promoted calibration, or sensitivity.
SCI-BEAM-REQ-012	The background family, origin, and coefficients shall be recorded; unsupported adaptive or higher-order backgrounds shall not be introduced.
SCI-BEAM-REQ-013	Admitted support shall record geometric selection, exclusions and causes, connectedness, edge/crop state, and retained sample count.
SCI-BEAM-REQ-014	The objective shall be Equation 7 with declared covariance or typed approximation; silent weighting or regularization is forbidden.
SCI-BEAM-REQ-015	Residuals shall mean admitted data minus the complete realized source-plus-background model on the retained support.
SCI-BEAM-REQ-016	Local identifiability shall be assessed for the realized support and model; non-identifiable parameters shall be unavailable with causes.
SCI-BEAM-REQ-017	Fit covariance shall identify its method and assumptions and shall satisfy Equation 8 only where those assumptions hold.
SCI-BEAM-REQ-018	Material source, coordinate, response, bandpass, atmosphere, and calibration nuisance uncertainties shall be propagated by a declared method or named unavailable.
SCI-BEAM-REQ-019	A covariance derived under an approximation shall be labeled conditional and shall not be described as calibrated by algebra alone.
SCI-BEAM-REQ-020	Response interpretation shall preserve the upstream response/kernel state; an incomplete state shall make intrinsic-beam interpretation unavailable.
SCI-BEAM-REQ-021	A prior shall record producer/version, array/network/slot identity, frame, unit, location, scale/covariance, reliability, compatibility, and fallback state.
SCI-BEAM-REQ-022	A compatible prior may initialize and bound locator gating but shall not enter the measurement objective as exact truth or veto the blind route.

ID	Requirement
SCI-BEAM-REQ-023	Missing, incompatible, weak, or unsuccessful prior use shall invoke the declared blind locator fallback and record the selected route.
SCI-BEAM-REQ-024	Candidate comparison shall use a common realized measurement model/objective so that a stronger observed compatible source can defeat a wrong soft prior.
SCI-BEAM-REQ-025	Prior influence shall report candidate route, displacement from prior, whether gating altered the searched domain, and whether the selected candidate came from prior or blind search.
SCI-BEAM-REQ-026	Locator and measurement phases, selected-candidate identity, parameter availability, support, valid-detector set, model identity, and their changes shall be recorded as one observation-local estimator trace.
SCI-BEAM-REQ-027	Convergence shall use all separately declared components of Equation 11, including per-parameter metrics, modulo-π angular distance, availability equality, objective, candidate, support, and valid-detector stability; optimizer completion alone shall not imply convergence.
SCI-BEAM-REQ-028	Alternating candidates, changed parameter availability, violated parameter/objective/support/valid-detector stability, and maximum-iteration termination shall have distinct non-converged causes.
SCI-BEAM-REQ-029	Every attempted detector shall have an attempt record and exactly one terminal disposition with cause codes.
SCI-BEAM-REQ-030	Parameter admissibility shall require finite values where defined, positive ordered widths, a valid normalization, and parameters within the declared model domain.
SCI-BEAM-REQ-031	BEAM-specific QC shall separate quantitative validity conditions from review cues and shall preserve all triggered causes rather than collapse them into one flag.
SCI-BEAM-REQ-032	Any numerical QC, convergence, support, or S/N threshold shall be a versioned policy input with owner and requested/effective/realized values; v0.1 defines no production defaults.
SCI-BEAM-REQ-033	A signal-to-noise quantity shall name its numerator, uncertainty denominator, covariance assumptions, and status; a formal ratio shall not be called empirical significance without evidence.
SCI-BEAM-REQ-034	Model inadequacy, low S/N, edge truncation, masked core, disconnected support, nonphysical parameters, prior domination, and non-convergence shall remain distinguishable states or causes.
SCI-BEAM-REQ-035	A typed calibration candidate may be formed only by Equation 10 with compatible source/amplitude conventions and declared uncertainty/dependence status.
SCI-BEAM-REQ-036	A calibration candidate shall carry source, amplitude, response, unit, covariance, observation, detector, model, and validity identities and shall not claim SCI-CAL promotion.
SCI-BEAM-REQ-037	Sensitivity shall be unavailable as a BEAM-owned v0.1 result; carried downstream diagnostics shall preserve their external authority and convention.
SCI-BEAM-REQ-038	Publication shall be atomic across result, QC, identity, policy, uncertainty, prior, convergence, candidate, and provenance records.
SCI-BEAM-REQ-039	Failure of a required bundle member shall fail publication; an optional map or TOD companion shall not substitute for it.
SCI-BEAM-REQ-040	Optional fit, APT, map, and TOD views shall carry explicit parent and compatibility identities and preserve BEAM state and limitations.
SCI-BEAM-REQ-041	Combining observations or using a result as a later prior shall require explicit compatibility checks and shall not mutate or retroactively redefine either observation-local result.
SCI-BEAM-REQ-042	Runs shall initialize all observation-local source, flux, prior, fit, convergence, QC, and validity state; execution order shall not affect an identical observation result.
SCI-BEAM-REQ-043	Duplicate rows, permuted rows, or repeated slot labels shall be resolved only through stable binding; ambiguous binding shall reject the affected relation.
SCI-BEAM-REQ-044	Equal numerical payloads with different validity or authority metadata shall remain scientifically distinct and shall not be silently merged.
SCI-BEAM-REQ-045	Outputs shall distinguish algebraic correctness, implementation conformance, response/representation fidelity, observational performance, and production readiness.
SCI-BEAM-REQ-046	Each claim unavailable because an upstream identity, covariance, response, calibration, or validity dependency is open shall be represented explicitly and causally.

8 Falsifiable predictions and edge cases

These predictions test the contract, not current software or production readiness.

ID	Predicted invariant or state
SCI-BEAM-PRED-001	Noiseless data generated by the admitted circular model on identifying support recover A as the excess at the translated declared reference origin, the zero-first-moment reference centroid, and common width; orientation is unavailable and objective residual is zero to numerical precision.
SCI-BEAM-PRED-002	Noiseless data generated by a non-circular admitted ellipse recover the translated zero-first-moment reference centroid, reference-origin amplitude, ordered widths, and θ modulo π on identifying support.
SCI-BEAM-PRED-003	Swapping unconstrained axis labels and adding $\pi/2$ leaves the profile invariant, but canonicalization restores major/minor ordering and the same canonical angle.
SCI-BEAM-PRED-004	Adding π to orientation leaves the model invariant; reported angles canonicalize to $[0, \pi)$.
SCI-BEAM-PRED-005	A constant background is recovered without changing source parameters for noiseless identifying support when the constant family is realized.
SCI-BEAM-PRED-006	A planar background is recovered under a realized planar family only when support identifies its gradients; fitting a constant to a true gradient produces structured residual/model-inadequacy evidence rather than an authorized gradient estimate.
SCI-BEAM-PRED-007	As finite source extent tends to zero while preserving the declared reference origin, the convolved $T(\mathbf{0}) = 1$ template tends to the point-source Gaussian; estimates change continuously where the problem remains identifiable.
SCI-BEAM-PRED-008	As source extent approaches beam scale, beam/source nuisance covariance grows or identifiability is lost; uncertainty shall reflect the supplied source-model uncertainty or the intrinsic-width claim becomes unavailable.
SCI-BEAM-PRED-009	Cropping or edge truncation can bias or de-identify centroid/shape; the result records edge state and cannot be valid merely because the optimizer converged.
SCI-BEAM-PRED-010	A masked core can remove reference-origin amplitude or width identifiability; those parameters become unavailable or rejected with a masked-core cause.
SCI-BEAM-PRED-011	Disconnected support is not silently joined; it is either admitted under an explicit support rule with identifiability demonstrated or rejected with cause.
SCI-BEAM-PRED-012	Zero admitted support yields no objective or fitted parameter and terminates as invalid-input or unavailable, never as a valid zero-amplitude fit.
SCI-BEAM-PRED-013	A compatible correct soft prior may change locator work but, under the same selected basin and measurement support, does not change the prior-free measurement optimum.
SCI-BEAM-PRED-014	Broad, weak, missing, incompatible, or unsuccessful priors select the declared blind route and cannot fabricate prior success.
SCI-BEAM-PRED-015	When a wrong compatible prior conflicts with a stronger observed candidate included by the blind route, common-objective comparison selects the better data-supported candidate and records the conflict.
SCI-BEAM-PRED-016	Permuting detector rows leaves stable-identity results invariant; duplicate or ambiguous bindings are rejected rather than resolved by position.
SCI-BEAM-PRED-017	Repeated slot labels do not merge detectors when stable occurrence-scoped bindings differ.
SCI-BEAM-PRED-018	Missing or incomplete response/kernel state preserves fitted effective-profile parameters where otherwise admissible but makes an intrinsic detector-plus-telescope beam claim unavailable.
SCI-BEAM-PRED-019	With correlated noise and its correct covariance, repeated standardized parameter errors follow the declared likelihood calibration; a deliberately diagonal approximation may preserve a point estimate yet its uncertainty remains conditional and can be miscalibrated.

ID	Predicted invariant or state
SCI-BEAM-PRED-020	A sequence alternating between selected candidate identities never satisfies the separately recorded candidate-stability component and terminates with an alternating-candidate or maximum-iteration cause, not valid convergence.
SCI-BEAM-PRED-021	Optimizer completion with an unstable per-parameter metric, availability state, objective, candidate, support, or valid-detector set is non-converged. Angles straddling 0 and π are compared by d_π, angle unavailable at circularity is omitted only when unavailable at both transitions, and exhausting $N_{\max}$ records maximum-iteration termination.
SCI-BEAM-PRED-022	Repeating identical observations in different run orders yields identical observation-local results and identities apart from execution provenance; no scientific state leaks from preceding runs.
SCI-BEAM-PRED-023	Candidate-factor uncertainty changes when source-flux or amplitude uncertainty changes; if their dependence is unknown, the candidate uncertainty is unavailable rather than computed as independent.
SCI-BEAM-PRED-024	Identical numerical payloads paired with different validity, response, calibration, or authority states remain different scientific results and cannot support the same downstream claim by payload equality alone.

9 Conformance report minimum

A conformance report shall give, for every requirement and prediction, an evidence identifier, applicable/not-applicable decision with reason, realized policy identity, observed result, expected result, and disposition. It shall list unavailable upstream claims and shall not represent a compile, test pass, synthetic match, or clean log as proof of a later authority gate.